

Md Shahid Ali Khan

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PROFILE SUMMARY

Backend Software Engineer with **3+ years of experience** in **Java and Spring Boot**, building **microservices and distributed systems** using **Kafka, Redis, REST APIs, concurrency, and asynchronous processing** in the **financial services domain**.

TECHNICAL SKILLS

Languages & Core CS: Java, Data Structures & Algorithms, OOP, Operating Systems, DBMS

Backend & Distributed Systems: Spring Boot, Spring Data JPA, Hibernate, REST APIs, Microservices, Spring WebFlux, Kafka, Redis, Caching, Concurrency, Multithreading, CompletableFuture, Resilience4j, System Design

Databases, Infrastructure & Tools: MySQL, PostgreSQL, Docker, Kubernetes, Jenkins (CI/CD), Git

WORK EXPERIENCE

Societe Generale – Software Engineer

July 2023 – Present

Report Generation & Risk Analytics Platform | *Spring Boot, Microservices, Kafka, Redis*

Bangalore

- **Drove the development and modernization** of a financial risk analytics platform generating **NPV, GAP, and NIM reports**, evolving it from a **modular monolith** toward independently deployable microservices.
- Designed a **two-level caching architecture** using an in-memory **L1 cache** and **Redis-based L2 distributed cache**, reducing API response time from **10 seconds to under 2 seconds** for optimized report requests.
- Optimized **cache keys, TTLs, and eviction**, cutting report generation time from **2 minutes to 30 seconds**.
- Refactored long-running report generation into **Kafka-driven asynchronous workflows**, enabling **concurrent report processing** and improving API responsiveness.
- Implemented resilient **Kafka consumer workflows** with **retries, Dead Letter Topics (DLTs), and automated email notifications** for failed report batches.
- Processed **50K+ records per report execution** across financial risk analysis and reporting workloads.

Societe Generale – Software Developer Intern

March 2023 – July 2023

Rejected Transaction Impact Analysis & Script Migration

Bangalore

- Built a **Spring Boot**-based analysis platform to categorize rejected transactions by risk, processing **100K+ transaction records** using optimized SQL queries to accelerate remediation workflows.
- Migrated **70+ legacy Unix shell scripts** to Linux and refactored workflows, improving system compatibility, maintainability, and operational efficiency.

PROJECTS

Distributed Payment Gateway | *Spring Boot, Microservices, Kafka, Redis, PostgreSQL, Kubernetes, Resilience4j*

- Built a **distributed payment gateway** supporting **Card, UPI, Net Banking, and Wallet** through a Strategy-based adapter layer, benchmarked at **up to 10,000 TPS using JMeter** on Kubernetes.
- Designed a secure **card-data vault** using **AES-256 envelope encryption** and Spring Security Crypto, with controlled handling of sensitive payment data.
- Implemented **SAGA-based payment workflows, the Transactional Outbox pattern, with Redis SETNX-based idempotency** to prevent duplicate payment processing during retries.
- Built a **webhook delivery service** with **HMAC-SHA256 signing, exponential backoff, and DLQ-based replay**, with **Resilience4j circuit breakers** and failure-injection testing for resilience.
- Deployed services on **Kubernetes with HPA-based scaling** and added **Prometheus, Grafana, and Zipkin** for metrics, dashboards, and distributed tracing.

EDUCATION

BIT Mesra

Ranchi, India

B.Tech in Information Technology | GPA: 8.60/10

2019 – 2023

- Received **“Spot” and “Newbie” Awards** at Societe Generale for strong performance and contributions.
- Secured **National Rank 844** in CODEKAZE 2021 among **200,000+ participants**.
- Scored in the **top 1 percentile** in JEE Main among **1 million+ candidates**.
- Awarded the merit-based **GP Birla Scholarship** at BIT Mesra, covering **50% of tuition fees**.